

UTKARSH SINGH

B.Tech Electronics & Communication Engineering | Jaypee Institute of Information Technology (JIIT), Noida
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ENGINEERING PROFILE

Full-stack developer, cybersecurity enthusiast, and ECE engineering student building secure software, interactive cybersecurity simulation platforms, ESP32 IoT systems, and scalable web applications.

CORE SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Framer Motion

Backend & APIs: Node.js, Express, FastAPI, Flask, REST APIs, JWT Authentication

Databases & Tools: PostgreSQL, MongoDB, Docker, Git, GitHub, Linux, Postman

Cybersecurity & IoT: Web Security, Stored XSS, ESP32, Sensor Telemetry, Hardware Integration

FEATURED ENGINEERING PROJECTS

1. Sentinel - Interactive Cybersecurity Learning Platform (Next.js, TypeScript, Tailwind, Gemini API)
Guided vulnerability labs and real-time attack/defense simulations for cybersecurity education.
2. XSS-Guard-Lab - Stored XSS Security Lab (Python, Flask, Web Security, OWASP)
Educational cybersecurity lab demonstrating Stored XSS attack vectors and secure remediation.
3. Sentinel Edge - ESP32 IoT Monitoring System (ESP32, Python, FastAPI, Next.js)
Hardware-to-cloud environmental telemetry pipeline streaming live sensor metrics to a web dashboard.
4. EdgeKart - Embedded Systems Marketplace (React, Node.js, Express, MongoDB, JWT)
Full-stack MERN e-commerce platform for electronics components with auth and admin dashboard.
5. PortFlow - Customs Broker Onboarding Platform (React, TypeScript, FastAPI, PostgreSQL, Docker)
Production-inspired multi-stage customer onboarding and compliance tracking workflow platform.
6. SafeSteps - Digital Safety Education Platform (React, Firebase, Google Maps, Tailwind CSS)
Interactive platform teaching digital safety with emergency assistance workflows.

EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida

B.Tech in Electronics & Communication Engineering (2025 - Present)